

RMC Mix Design Collection – M25, M30 & M35

Multiple grades and alternative formulations designed to create meaningful retrieval differences for an RMC RAG system.

M25 Concrete – Mix Design M25-A

Application: Residential and commercial RCC **Target Slump:** 90 mm

Material / Parameter	Quantity	Unit	Remarks
Cement	340	kg/m ³	Primary cementitious binder
Fly Ash	60	kg/m ³	Supplementary cementitious material
Water	172	L/m ³	Before aggregate moisture correction
Fine Aggregate / Sand	740	kg/m ³	Controlled fine aggregate
10 mm Coarse Aggregate	430	kg/m ³	Nominal coarse aggregate fraction
20 mm Coarse Aggregate	650	kg/m ³	Nominal coarse aggregate fraction
Chemical Admixture	3.5	kg/m ³	Water-reducing admixture
Water/Cementitious Ratio	0.43	-	Illustrative design value
Target Mean Strength	31.6	MPa	Illustrative 28-day target

Design remarks: Standard illustrative M25 formulation with moderate cementitious content.

M25 Concrete – Mix Design M25-B

Application: RCC requiring higher workability **Target Slump:** 110 mm

Material / Parameter	Quantity	Unit	Remarks
Cement	375	kg/m ³	Primary cementitious binder
Fly Ash	45	kg/m ³	Supplementary cementitious material
Water	165	L/m ³	Before aggregate moisture correction
Fine Aggregate / Sand	720	kg/m ³	Controlled fine aggregate
10 mm Coarse Aggregate	420	kg/m ³	Nominal coarse aggregate fraction
20 mm Coarse Aggregate	645	kg/m ³	Nominal coarse aggregate fraction
Chemical Admixture	4.1	kg/m ³	Water-reducing admixture
Water/Cementitious Ratio	0.393	-	Illustrative design value
Target Mean Strength	31.6	MPa	Illustrative 28-day target

Design remarks: Alternative M25 design with higher cement content and reduced water.

M30 Concrete – Mix Design M30-A

Application: Commercial building structural concrete **Target Slump:** 105 mm

Material / Parameter	Quantity	Unit	Remarks
Cement	380	kg/m ³	Primary cementitious binder
Fly Ash	80	kg/m ³	Supplementary cementitious material
Water	170	L/m ³	Before aggregate moisture correction
Fine Aggregate / Sand	720	kg/m ³	Controlled fine aggregate
10 mm Coarse Aggregate	420	kg/m ³	Nominal coarse aggregate fraction
20 mm Coarse Aggregate	650	kg/m ³	Nominal coarse aggregate fraction
Chemical Admixture	4.2	kg/m ³	Water-reducing admixture
Water/Cementitious Ratio	0.37	-	Illustrative design value
Target Mean Strength	38.25	MPa	Illustrative 28-day target

Design remarks: Illustrative M30 design with fly ash and chemical admixture.

M30 Concrete – Mix Design M30-B

Application: High-workability structural concrete **Target Slump:** 120 mm

Material / Parameter	Quantity	Unit	Remarks
Cement	410	kg/m ³	Primary cementitious binder
Fly Ash	60	kg/m ³	Supplementary cementitious material
Water	165	L/m ³	Before aggregate moisture correction
Fine Aggregate / Sand	705	kg/m ³	Controlled fine aggregate
10 mm Coarse Aggregate	415	kg/m ³	Nominal coarse aggregate fraction
20 mm Coarse Aggregate	645	kg/m ³	Nominal coarse aggregate fraction
Chemical Admixture	4.7	kg/m ³	Water-reducing admixture
Water/Cementitious Ratio	0.351	-	Illustrative design value
Target Mean Strength	38.25	MPa	Illustrative 28-day target

Design remarks: Alternative M30 formulation with higher cement content and lower water.

M35 Concrete – Mix Design M35-A

Application: Heavy structural applications **Target Slump:** 110 mm

Material / Parameter	Quantity	Unit	Remarks
Cement	400	kg/m ³	Primary cementitious binder
Fly Ash	70	kg/m ³	Supplementary cementitious material
Water	165	L/m ³	Before aggregate moisture correction
Fine Aggregate / Sand	690	kg/m ³	Controlled fine aggregate
10 mm Coarse Aggregate	410	kg/m ³	Nominal coarse aggregate fraction
20 mm Coarse Aggregate	650	kg/m ³	Nominal coarse aggregate fraction
Chemical Admixture	4.6	kg/m ³	Water-reducing admixture
Water/Cementitious Ratio	0.351	-	Illustrative design value
Target Mean Strength	43.25	MPa	Illustrative 28-day target

Design remarks: Illustrative M35 mix with controlled water and increased cementitious content.

Dataset note: These are synthetic RMC mix-design records created for RAG ingestion, retrieval, comparison, and question-answering tests. They are not approved construction designs. Actual mix proportions must be established and validated using project requirements, material properties, laboratory trials, moisture corrections, and applicable standards.